



DECEMBER 2021: END SEMESTER ASSESSMENT (ESA) B TECH 6th SEMESTER

UE17/18CS351 - Compiler Design

Time: 3 Hrs	Answer All Questions	Max Marks: 100
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1	a)	Define Compiler. What are the phases of the Compiler? Explain with a neat diagram. Mention the input and output for each phase.	10																					
	b)	What is Transition Diagram? Draw transition diagram that recognize the lexeme matching the token 'relop'. The attribute values are mentioned in the following table. <table><tr><td>Lexeme</td><td>Token</td><td>Attribute Value</td></tr><tr><td><</td><td>relop</td><td>LT</td></tr><tr><td><=</td><td>relop</td><td>LE</td></tr><tr><td>=</td><td>relop</td><td>EQ</td></tr><tr><td><></td><td>relop</td><td>NE</td></tr><tr><td>></td><td>relop</td><td>GT</td></tr><tr><td>>=</td><td>relop</td><td>GE</td></tr></table>	Lexeme	Token	Attribute Value	<	relop	LT	<=	relop	LE	=	relop	EQ	<>	relop	NE	>	relop	GT	>=	relop	GE	4
	Lexeme	Token	Attribute Value																					
	<	relop	LT																					
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>	relop	GT																						
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c)	Identify tokens generated by Lexical Analyzer for the following code. for (int k = 0; k < myArray[5]; ++k) { cout << k << endl; }	2																						
d)	Explain with the neat diagram, input buffering with the use of sentinels. Also write algorithm/pseudo-code for look-ahead code with sentinels.	4																						
2	a)	Consider the following CFG, which has the set of terminals T = {id , (,) , [,] , ; } E → id id(A) id [E] A → E E ; A (a) Left-factor this grammar. (b) Write FIRST and FOLLOW sets. (c) Construct an LL(1) parsing table for the left-factored grammar. (d) Show the operation of an LL(1) parser on the input string id(id[id]; id)	10																					
	b)	What is LR(0) item? Explain with example. Consider the grammar: S → AA A → aA b Let I is { [S' → .S] }, compute the set CLOSURE(I).	3																					

	c)	Consider the grammar $E' \rightarrow E$ $E \rightarrow E+T \mid T$ $T \rightarrow T * F \mid F$ $F \rightarrow (E) \mid id$ Let $I = \{ E' \rightarrow .E, E \rightarrow .E+T, E \rightarrow .T, T \rightarrow .T * F, T \rightarrow .F, F \rightarrow .(E), F \rightarrow .id \}$ Compute $goto(I, X)$ for all possible values of X .	2
	d)	Consider following Grammar. Construct LR(0) automaton, SLR parsing table. $S \rightarrow (L) \mid a$ $L \rightarrow L, S \mid S$	5
3	a)	The grammar to translate binary integer number into decimal is given below. Let synthesized attribute $B_N.val$ gives the value of decimal equivalent of binary number (eg. For input 11001, $B_N.val = 25$). $B_N \rightarrow L$ $L \rightarrow L_1 B$ $L \rightarrow B$ $B \rightarrow 0$ $B \rightarrow 1$ (a) Give translation scheme for above grammar. (b) Draw annotated parse tree for '10111'	8
	b)	With the examples, explain in detail the different types of attributes used to define the semantic rules.	5
	c)	What is dependency graph? Explain with example. Draw the dependency graph for the string '10111' using the translation scheme for question 3(a).	7
4	a)	Provide Three-address code and its Quadruple, Triple and Indirect Triple Representation for $x = y[i] + z[j]$	7
	b)	What is directed acyclic graph? Construct Syntax tree and DAG for the following expression (use unambiguous expression grammar for reference) $[(a+b)*c+((a+b)+e)*(e+f)]*[(a+b)*c]$	6

	c)	What is basic block and flow graph? Partition following sequence of three-address instructions into basic blocks. Also construct flow graph. 1. $i=1$ 2. $j=1$ 3. $t1=10*i$ 4. $t2=t1+j$ 5. $t3=8*t2$ 6. $t4=t3-88$ 7. $a[t4]=0.0$ 8. $j=j+1$ 9. If $j \leq 10$ goto (3) 10. $i=i+1$ 11. If $i \leq 10$ goto (2) 12. $i=1$ 13. $t5=i-1$ 14. $t6=88*t5$ 15. $a[t6]=1.0$ 16. $i=i+1$ 17. If $i \leq 10$ goto(13)	7
5	a)	Explain different issues involved in the design of a code generator.	8
	b)	Explain the structure and each field of a general activation record with the diagram.	6
	c)	Explain access link with example.	3
	d)	Explain calling sequence and division of tasks between caller and callee.	3